# Bachelor's Thesis Assignment

169976

Institut: Department of Intelligent Systems (DITS)
Student: **Kapitulčín Jakub**
Programme: Information Technology
Title: **An environment for the visual design, monitoring, and management of distributed control systems and IoT networks**
Category: Software Engineering
Academic year: 2025/26

Assignment:

1. Study the principles of developing distributed control systems. Familiarize yourself with the DEVS formalism. Familiarize yourself with the IEC-61499 standard. Familiarize yourself with the NodeRED and Eclipse 4diac/FORTE tools, as well as the ROS2 and MicroROS environments.
2. Design a runtime environment (RTE) capable of interpreting IEC-61499-inspired communicating, interpreted, timed automata on various platforms in a distributed environment. Enable access to sensors and actuators connected to GPIO and communicating via protocols such as I2C, etc.
3. Design a development environment (IDE) and related infrastructure enabling: (1) visual modeling of distributed control systems based on hierarchical block diagrams in the DEVS (Discrete Event Systems Specification) style and interpreted timed automata in the IEC-61499 style, (2) code generation for execution, (3) its deployment to target devices based on the distributed system model, (4) monitoring of system execution, (5) time travel debugging, (6) dynamic reconfiguration of the running system. The infrastructure should enable interoperability with third-party devices (WiFi relays, Zigbee sensors, etc.) as well as off-the-shelf subsystems based on ROS2, etc.
4. Implement the RTE so that it can run on microcontrollers such as ESP, RPI Pico, etc., as well as on computers running Linux and the ROS2 system. For communication, consider MQTT, UDP, and DDE. Implement the IDE as a web application.
5. Demonstrate the functionality and features of the implemented system using a suitable example of a distributed control system.

Literature:
- Dle pokynů vedoucího práce.

Detailed formal requirements can be found at https://www.fit.vut.cz/study/theses/

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